

JEE PYQs

You may have already seen these questions before, but nonetheless repeated practice is good. ☺

1. (JEE Main 2024) A particle starts from rest and velocity varies as $v = 4\sqrt{x}$. Find its acceleration.
2. (JEE Main 2020) Given $x^2 = at^2 + 2bt + c$, acceleration depends on x as x^{-n} . Determine integer n .
3. (JEE Main 2019) Position $x(t) = at + bt^2 - ct^3$. At zero acceleration, find velocity.
4. (JEE Main 2021) Velocity $v = \alpha t + \beta t^2$. Compute displacement between $t = 1\text{s}$ and $t = 2\text{s}$.
5. (JEE Main 2021) Force as a function of time is $F(t) = F_0 \left[1 - \left(\frac{t-T}{T} \right)^2 \right]$ acts for $2T$. Find velocity after $2T$. (also given in a previous assignment)
6. (JEE Advanced 2022) For $\mathbf{F} = -k(x\mathbf{i} + y\mathbf{j})$, $k = 1$, at $z = 0.5$, compute $(xv_y - yv_x)$.
7. (JEE Main 2021) Velocity is related to time as $v = \alpha t + \beta t^2$. Derive displacement via integration.
8. (JEE Main 2021) Particle on horizontal surface has a retarding force $F = -kx$. Find exit speed after patch of length L assuming initial speed was v_i

Additional questions

11. A particle under $F(x) = kx^2$ starts from rest at $x = 0$. Find $v(x)$.
12. Acceleration $a = -\alpha v^2$, with $v(0) = v_0$. Derive $v(t)$.
13. Position $x(t) = A \sin(\omega t)$. Find force on mass m . (this concept comes later in waves and oscillations)
14. Body thrown with drag force $F_d \propto v$. Derive terminal velocity and $v(t)$. (this will be discussed further in fluid mechanics where you learn the stoke's law for a spherical body)
15. $x(t) = at^3 - bt^2$. Find acceleration as function of x .
16. Variable mass problem: in a system, the mass changes with velocity as $\frac{dm}{dt} = bv$. Derive velocity vs time. (to convince yourself that this is possible, think of a train with a compartment carrying sand, which on increment in velocity results in sand falling out and the net mass decreasing)
17. Given $U(x) = Ax^4$, find force $F(x)$, and if total energy is E , find speed as function of x .
18. Projectile with $a(v) = -kv$. Solve for $v(t)$.
19. Friction $F(x) = \mu mg e^{-x/L}$. Compute work from 0 to L . (as mentioned in a previous assignment, work done is simply $W = \int_{x_i}^{x_f} F dx$ where x_i and x_f are initial and final positions.)

20. Force $F(t) = F_0 e^{-kt}$ on mass m , starting from rest—find $v(t)$. (use the integral relation for exponential function, $e^x dx = e^x$)